

6.10 Solution: The torque on all the particles in a volume V is

$$\frac{dL_{\text{mech}}}{dt} = \int_V \mathbf{x} \times (\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) d^3x.$$

Following the derivation in Section 6.7,

$$\begin{aligned} \mathbf{x} \times (\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) &= \mathbf{x} \times \left[\epsilon \mathbf{E}(\nabla \cdot \mathbf{E}) - \epsilon \mathbf{E} \times (\nabla \times \mathbf{E}) + \mu \mathbf{H}(\nabla \cdot \mathbf{H}) \right. \\ &\quad \left. - \mu \mathbf{H} \times (\nabla \times \mathbf{H}) \right] - \mu \epsilon \mathbf{x} \times \frac{\partial}{\partial t}(\mathbf{E} \times \mathbf{H}). \end{aligned}$$

Then, we can define the field angular momentum density as

$$\mathcal{L}_{\text{field}} = \mu \epsilon \mathbf{x} \times (\mathbf{E} \times \mathbf{H}),$$

and the conservation of angular momentum becomes

$$\begin{aligned} \frac{d}{dt}(L_{\text{mech}} + L_{\text{field}}) &= \frac{d}{dt} \int_V (\mathcal{L}_{\text{mech}} + \mathcal{L}_{\text{field}}) d^3x \\ &= \int_V \mathbf{x} \times \left(\epsilon \mathbf{E}(\nabla \cdot \mathbf{E}) - \epsilon \mathbf{E} \times (\nabla \times \mathbf{E}) + \mu \mathbf{H}(\nabla \cdot \mathbf{H}) - \mu \mathbf{H} \times (\nabla \times \mathbf{H}) \right) d^3x \\ &= \epsilon_{ijk} \hat{e}_i \int_V x_j \frac{\partial}{\partial x_l} T_{kl} d^3x, \end{aligned}$$

where we have used Einstein summation convention. Notice that

$$\begin{aligned} \epsilon_{ijk} x_j \frac{\partial}{\partial x_l} T_{kl} &= \epsilon_{ijk} \left[\frac{\partial}{\partial x_l} (x_j T_{kl}) - T_{kl} \frac{\partial}{\partial x_l} x_j \right] \\ &= \frac{\partial}{\partial x_l} (\epsilon_{ijk} x_j T_{kl}) - \epsilon_{ijk} \delta_{jl} T_{kl} \\ &= \frac{\partial}{\partial x_l} (\epsilon_{ijk} x_j T_{kl}) - \epsilon_{ijk} T_{kj}, \end{aligned}$$

where $\epsilon_{ijk} T_{kj}$ is zero, as $\overleftrightarrow{T}$ is a symmetric tensor. Therefore,

$$\frac{d}{dt} \int_V (\mathcal{L}_{\text{mech}} + \mathcal{L}_{\text{field}}) d^3x = \int_V \nabla \cdot (\mathbf{x} \times \overleftrightarrow{T}) = - \int_V \nabla \cdot \overleftrightarrow{M} d^3x,$$

with

$$\overleftrightarrow{M} = \overleftrightarrow{T} \times \mathbf{x}.$$

Now we can write the conservation of angular momentum in the integral form as

$$\frac{d}{dt} \int_V (\mathcal{L}_{\text{mech}} + \mathcal{L}_{\text{field}}) d^3x + \int_V \mathbf{n} \cdot \overleftrightarrow{M} da = 0,$$

and the corresponding differential form is

$$\frac{\partial}{\partial t} (\mathcal{L}_{\text{mech}} + \mathcal{L}_{\text{field}}) + \nabla \cdot \overleftrightarrow{M} = 0.$$